Class 5: Data Viz with ggplot2

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Background

There are many graphics systems in R for making plots and figures. These include so-called *"base-R"* graphics like the `plot()` function and add on packages like **ggplot2**.

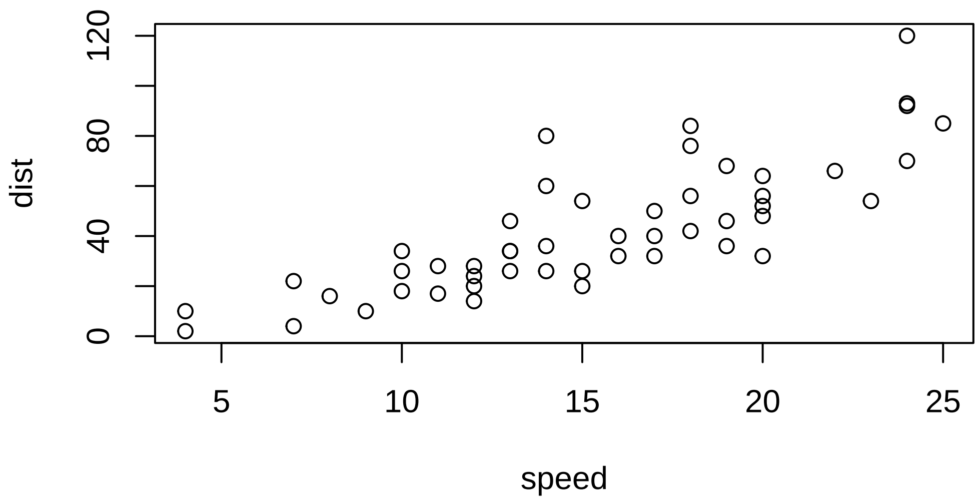
Let's compare how we make a simple figure with these two systems:

We can use the in-built `cars` dataset:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```
plot (cars)
```



Before I can use ggplot2 I need to install it on my computer. To do this we can use the function `install.packages("ggplot2")`

> **N.B.** We never run `instal.packages()` in our quarto doc (we run it once only in our R console) as it would re-install the package every time we render our quarto report.

Once installed we need to load up the package into our R brain:

```
library(ggplot2)
```

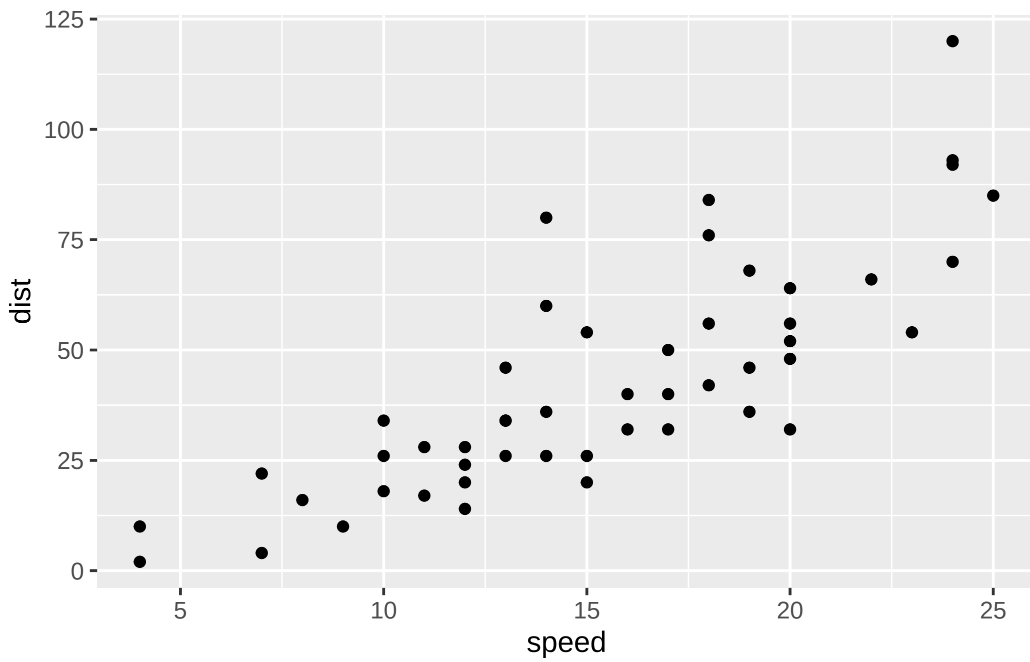
The main function in the **ggplot2** package is called `ggplot()`

```
ggplot(cars)
```

Every ggplot has at least 3 layers:

- the **data** (a.data.frame of the stuff we want to plot)
- the **aes**thetics (how the data maps to the plot),
- the **geom** layer () (how you want the plot to be drawn, e.g. points, lines, etc.)

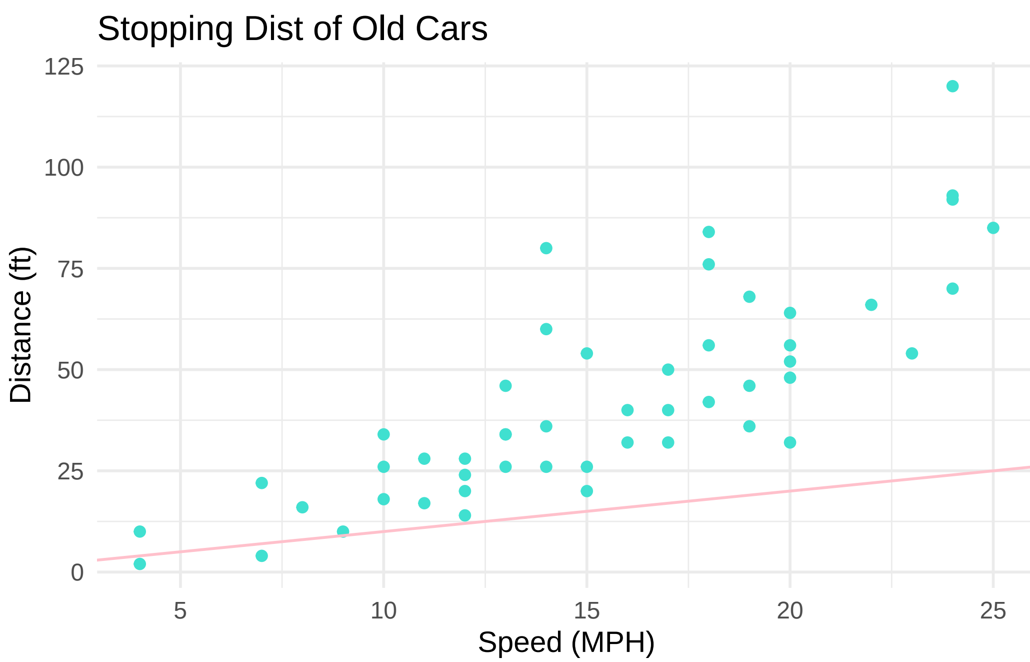
```
ggplot(cars) +
  aes(x=speed, y=dist ) +
  geom_point()
```



Add some custom features

Let's add a trend line that shows the relationship between speed and distance.

```
ggplot(cars) +
  aes(x=speed, y=dist ) +
  geom_point(color = "turquoise") +
  geom_abline (color = "pink") +
  theme_minimal() +
  labs(title="Stopping Dist of Old Cars",
       x = "Speed (MPH)",
       y = "Distance (ft)")
```



Q. Can you make the `geom_smooth()` function produce a linear straight line fit to the data and turn off the "gray" area region.

Gene Plot

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

```r
sum(genes$State == "up")
```

```
[1] 127
```

A useful new function in this context is the `table()` function:

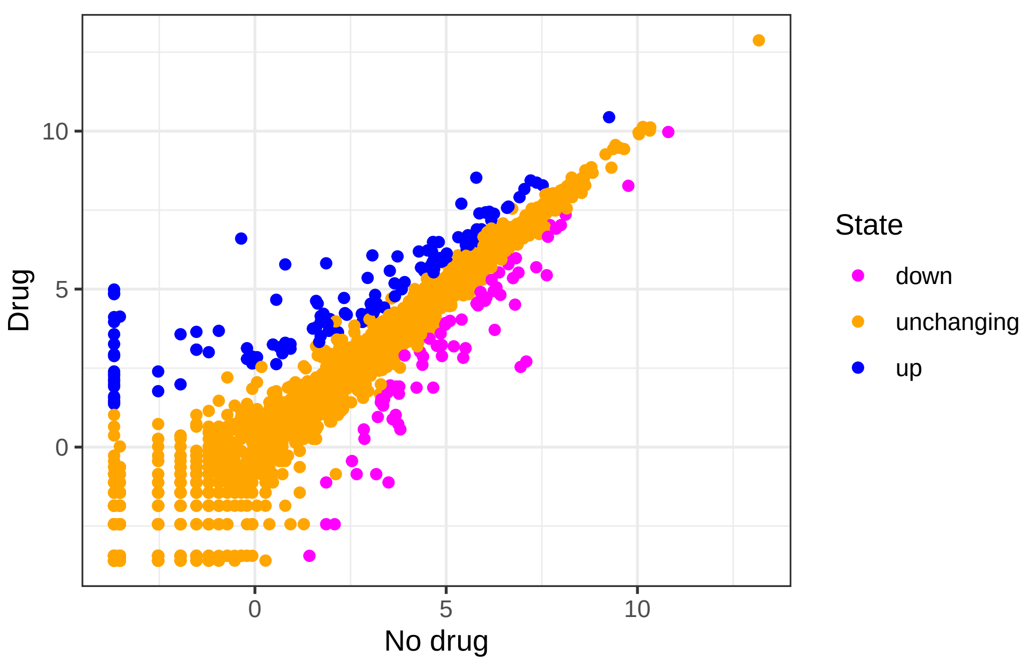
```r
table(genes$State)
```

```

      down unchanging         up 
        72       4997        127 
```

My first plot attempt

```r
ggplot(genes) +
  aes(Condition1, Condition2, col=State) +
  geom_point() +
  scale_color_manual(values=c("magenta","orange","blue" )) +
  theme_bw() +
  labs(x = "No drug", y="Drug")
```



Going further

Here we read the famous gapmider dataset:

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

> Q. How many entries (i.e. rows) are in this dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

> Q. How many different "country" entries are in this dataset?

```
length(unique(gapminder$country))
```

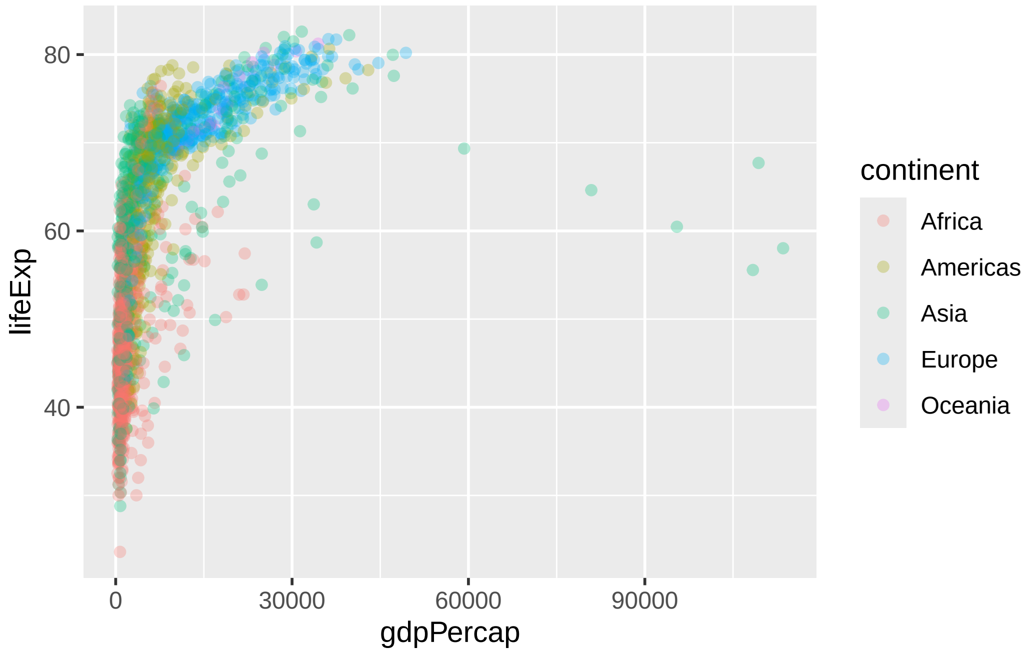
```
[1] 142
```

Let's make our first plot of the entire dataset:

Plot of "gdpPercap" vs "lifeExp" colored by "continent"

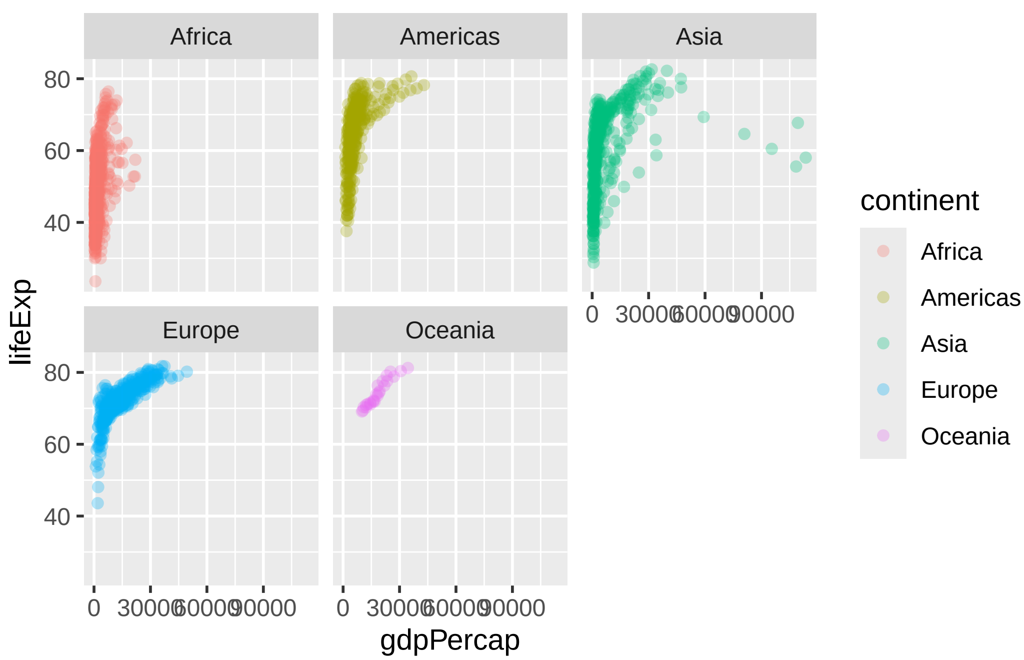
```
p <- ggplot(gapminder) +
  aes(gdpPercap,lifeExp, col=continent) +
  geom_point(alpha=0.3)
```

```
p
```



I can add more layers to `p`

```
p +
  facet_wrap(~continent)
```



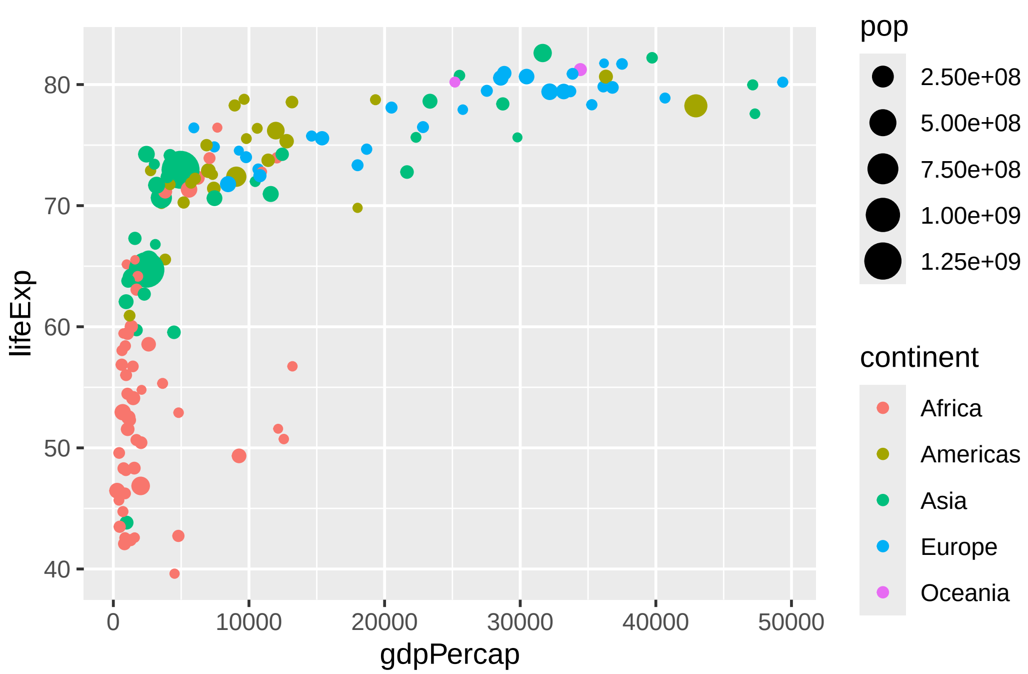
Make a plot for years 1977 and 2007 only (not all the years in the dataset).

> Q. First use the **dplyr** package and use the `filfter()` function from that package to extract the year 2007.

```
library (dplyr)
library (ggplot2)
```

```
g07 <- filter(gapminder, year== 2007)
g77 <-filter(gapminder, year== 1977)
```

```
ggplot(g07) +
  aes(gdpPercap,lifeExp, col=continent, size=pop) +
  geom_point()
```




```
  facet_wrap(~year)
```

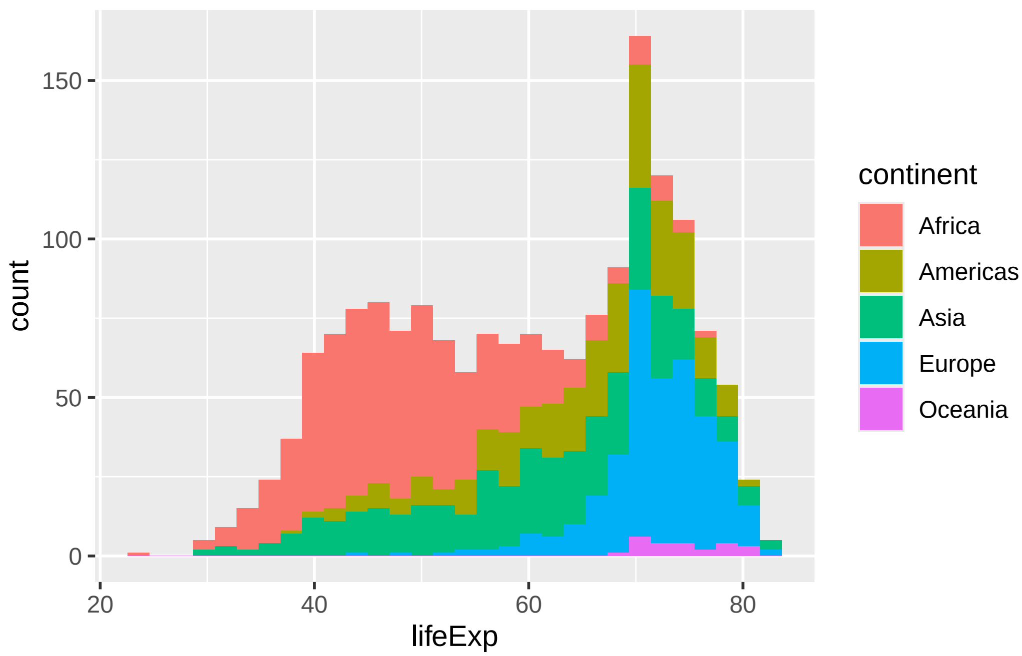
```
<ggproto object: Class FacetWrap, Facet, gg>
    attach_axes: function
    attach_strips: function
    compute_layout: function
```

```
    draw_back: function
    draw_front: function
    draw_labels: function
    draw_panel_content: function
    draw_panels: function
    finish_data: function
    format_strip_labels: function
    init_gtable: function
    init_scales: function
    map_data: function
    params: list
    set_panel_size: function
    setup_data: function
    setup_panel_params: function
    setup_params: function
    shrink: TRUE
    train_scales: function
    vars: function
    super:  <ggproto object: Class FacetWrap, Facet, gg>
```

Q. Make a histogram of lifeExp colored by continent (try using fill = or `col=continent` or `col=continent`)

```
ggplot(gapminder) +
    aes(lifeExp, fill= continent) +
    geom_histogram()
```

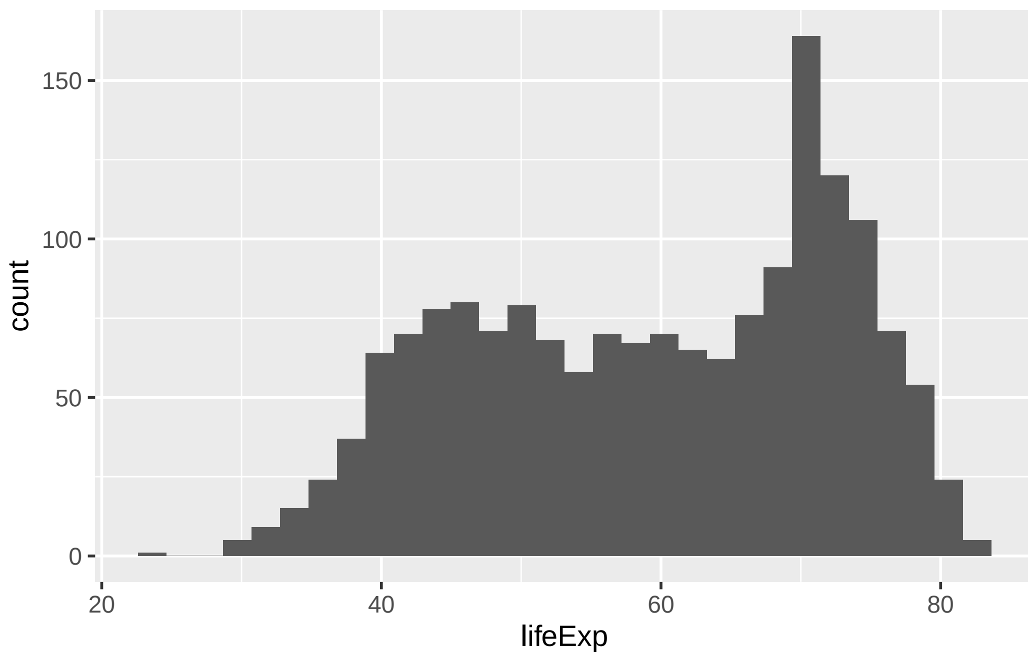
```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Q. Make a histogram of lifeExp faceted by continent

```
ggplot(gapminder) +
    aes(lifeExp) +
    geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



150
100
50
0
count
20
40
60
80
lifeExp